

Xiaojun Ge

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Relevant Research Skills

Field Experimentation

Hydrological monitoring device usage
On-site water quality monitoring

Multi-depth soil profile & coring sampling
Soil/Water GHG sampling (chamber), forest flux (tower)

Nutrient Transformation

- ^{15}N isotopic tracing method

- Sediment incubation

Soil Physicochemical Measurement

- Soil enzyme activity assays
- Soil aggregate and stability measurement

- Soil sesquioxide quantification
- Nutrient fractionation separation (C, N, P)

Microbial and Genetic Analysis

16S rRNA amplicon sequencing
qPCR (Quantitative Polymerase Chain Reaction)

QIIME2 analysis: phylogeny, QC & visualization
Functional gene annotation & analysis

Instrumental operation and maintenance

- Automated & continuous flow analyzers
- CRDS (Cavity Ring-Down Spectroscopy)
- HPLC-MS-MS

- UV-Vis spectroscopy
- Unmanned Aerial Vehicle

Software Skills

- R
- Python & PyTorch

- GIS & Google Earth Engine (GEE)

Research Experience

Research Assistant

Institute of Guangxi Zhuang Autonomous Region Institute of Metrology & Test, Nanning, China

Development and validation of a high-resolution carbon sink assessment system for plantations **Nov. 2022 - Present**

- Integrated multi-scale data with machine learning approaches to optimize carbon sink model parameters
- Assessed pathways for economic realization of plantation carbon sinks by integrating empirical analysis, Machine-learning-driven modeling, and policy evaluation, securing ¥200,000 project funding.

Joint-Training Graduate Student

Supervisor: Zaijian Yuan

Institute of Eco-environmental and Soil Sciences, Guangdong Academy of Science, China

Technology Integration and Demonstration of Manure Pollution Ecological Treatment in Swine Farms Project **Sep. 2020 – June 2022**

- Led the design and establishment of mesocosm constructed surface flow constructed wetlands .
- Participated in the design and construction of field-scale surface-flow constructed wetland.
- Teamed with group colleagues to refine wetland and execute isotope experiments
- Experimentally studied the impact of antibiotic exposure time on nitrogen gross transformation rates.
- Routinely monitored and quantitatively analyzed the impact of sustained antibiotic exposure on nitrogen removal efficiency and pathways in wetlands.
- Revealed changes in the microbial community of wetlands under long-term antibiotic exposure.

Nanling Forest National Field Observation and Research Station Project

May. 2020 – Dec. 2021

- Participated in establishing soil monitoring plots and collecting soil samples.
- Analyzed soil physicochemical properties along different altitudinal gradients and across various seasons.
- Investigated the relationships between carbon fractions, soil physicochemical properties, and the microbial community.

Hydrological and ecological responses to deep vertical rotary tillage (DVRT)

Sep. 2019 – June 2021

- Developed a portable rainfall simulation device for several hydrological projects.
- Designed a variety of experiments with team members.
- Implemented field rainfall simulations experimental with two graduate students for two months.
- Compared the soil stability, sesquioxides, and nutrients fraction composition of DVRT with traditional tillage.
- Quantitatively assessed the risk of soil erosion and nutrient loss under DVRT.
- Provided further understanding of the effect of DVRT on hydrological ecology.

Publications

1. Alleviating heavy metal migration risks by adding biochar during sludge amelioration of forest soil: Evaluating differential loss across multiple media. **(Third Author)** 2026. *Journal of Hazardous Materials*, 503, 141098.
DOI: <https://doi.org/10.1016/j.jhazmat.2026.141098>
2. Risk and driving factors of nitrogen and phosphorus runoff losses under extreme rainfall conditions: sludge application strategies in forest soils. **(Co-first Author)** 2025. *CATENA*, 259, 109318.
DOI: <https://doi.org/10.1016/j.catena.2025.109318>
3. Temporal characteristics and influencing factors of agricultural net carbon sink in Guangxi from 1978 to 2021. **(First Author)** 2024. *Chinese Journal of Eco-Agriculture*, 32(2), 218–229.
DOI: <https://doi.org/10.12357/cjea.20230523>
4. Effects of biochar application on the loss characteristics of Cd from acidic soil under simulated rainfall conditions. **(Sixth Author)** 2022. *Environmental Science and Pollution Research*, 29(55), 83969–83980.
DOI: <https://doi.org/10.1007/s11356-022-21623-x>
5. Temporal and spatial variation characteristics and source analysis of agricultural non-point source pollution load in Guangdong during the past 20 years. **(First Author)** 2022. *Environmental Science*, 43(6), 3118–3127.
DOI: <https://doi.org/10.13227/j.hjkx.202108071>
6. Effect of Short- and Long-Term Antibiotic Exposure on Microbially Mediated Nitrogen Reduction-Oxidation Process in Sediment of Surface-Flow Constructed Wetland. **(First Author)** *In preparation.*
7. Soil property-mediated effects of deep vertical rotary tillage on sheet erosion under simulated rainfall. **(First Author)** *In preparation*

Education & Honor

South of China Normal University, M.S. in Environmental Science & Engineering**Sep. 2019 – June 2022****Thesis:** Study on the effect and mechanism of antibiotics on nitrogen removal in constructed wetland in non-point source pollution of livestock treatment**Rewards:** 3 years of school-level scholarships of South of China Normal University (2019–2021)**Guangxi Normal University**, B.S. in Environmental Science**Sep. 2014 – June 2018**

References

Prof. Zaijian Yuan: Director and Professor, Department of Non-Point Source Pollution Control and Soil and Water Conservation, Institute of Eco-environmental and Soil Sciences, Guangdong Academy of Sciences**Email:** zjyuan@soil.gd.cn**Dr. Xianli Xu:** Professor, Institute of Subtropical Agriculture, CAS; Member of the IALE-China Council**Email:** xianlixu@isa.ac.cn**Dr. Jing Zhu:** Associate Professor, College of Environment and Resources, Guangxi Normal University**Email:** zhuj@gxnu.edu.cn